

Review retrospective:
*the harmonic cross-susceptibility capstone, and a docstring the
file outgrew*

Claude Opus 4.8 (1M ctx), reviewer GPT-5.5 Pro

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Abstract

A soundness review of `Laplace/OneD/HarmonicCrossSuscDeriv.lean`: the capstone composing the threepoint κ_3 machinery into a cross-susceptibility statement — for the harmonic Gibbs with linear perturbation, $\partial_h \text{Cov}_h(x, x)|_0 = -t \kappa_3(x, x, x) = 0$. Result: **a clean fidelity pass with two docstring-staleness fixes** — the module doc described an “external / future-candidate” state the file (and its sibling) had since outgrown.

1 Setting

A cross-seabed composition: it imports `Threepoint.Harmonic` for the FDT identity $\partial_h \text{Cov}_h(\varphi, B) = -t \kappa_3(\varphi, A, B)$ and the parity vanishing $\kappa_3(x, x, x) = 0$, and the laplace monomial-observability lemmas to make the result unconditional. It is the first reviewed file that bridges laplace and threepoint.

2 What was reviewed

Two theorems: the conditional `cov_h_id_id_deriv_harmonic_eq_zero` (taking three `GibbsObservable` hypotheses) and its `_unconditional` variant.

3 Fidelity / soundness findings

Sound; no proof defect. The composition is exactly the intended chain: regularity instance $\rightarrow$ FDT identity $\rightarrow$ rewrite $\kappa_3 = 0 \rightarrow \text{mul_zero}$. The reviewer confirmed the six `GibbsObservable` slots of the derivative theorem collapse correctly to the three caller-facing hypotheses under $A = B = \varphi = \text{id}$ (the `hx hx hx2 hx2 hx2 hx3` threading), and that the unconditional variant discharges exactly those three.

Two stale module-docstring status claims, fixed. (1) The doc said the monomial `GibbsObservable` facts “remain external ... not formalised here” — but the `_unconditional` theorem *in this same file* discharges them; corrected. (2) The doc said the stronger global-constancy theorem ($\text{Cov}_h(x, x) = 1/(\lambda t)$ for all h) was a “future tide candidate” — but it is now proved in `HarmonicCovGlobalConstancy.lean` (`gibbsCov_h_id_id_harmonic_eq`, verified); corrected to point there. Both comment-only. A third drift — inconsistent tide numbering (10 / 11 / 13) in the provenance prose — was left report-only, as the numbering is unreconcilable from the file alone and the sibling file uses the same “Tide 13” label.

4 Simplifications applied

None beyond the `docstring` fixes — the simplification bucket was empty. `open MeasureTheory` is load-bearing (bare `Measure  $\mathbb{R}$ /volume`); there is no `open Real`; all three imports are used; no no-op tactic. GPT affirmed.

5 Disagreements and open questions

None.

6 Lessons

- **A file can outgrow its own module `docstring`.** The doc was written when only the conditional theorem existed and the global-constancy fact was hypothetical; the unconditional theorem and the sibling file landed later, leaving the “external / future candidate” prose false. Status claims (“not formalised here”, “future candidate”) age the fastest — check them against the file’s *current* contents and its siblings, and (for a cross-reference) by actually opening the named target.
- **Verify a forward-reference before writing it in.** Before pointing the `docstring` at `HarmonicCovGlobalConstancy.lean`, the review confirmed that file’s `gibbsCov_h_id_id_harmonic_eq` actually states the global identity — a fixed cross-reference is only an improvement if it resolves.

7 Follow-ups

- `HarmonicCovGlobalConstancy.lean` (427L) is the natural next review — the global-constancy theorem this file now defers to.
- The remaining `OneD/ Harmonic*/Anharmonic*` files and `IntegralRemainder.lean` (2668L, the asymptotic-remainder foundation).